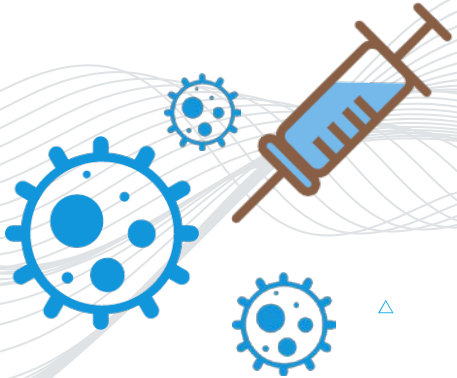
全球新冠疫苗接受度与信心监测

——基于社交媒体与深度学习的研究

挑战杯市赛答辩

指导老师：侯志远副教授
答辩人：周新宇



研究背景

新冠疫情泛滥，控制疫情的希望在于疫苗接种，然而疫苗迟疑广泛存在，影响接种率。**各国应该根据新冠疫苗接种意愿制定接种计划。**

现有研究的不足

1）**大多数调查局限在少数国家**，迄今最大的全球新冠疫苗犹豫研究仅比较了**32国**，**许多国家被现有研究忽视**；

2）研究显示在部分国家，新冠疫苗接受度在不断变化；然而，全球、各国、各区域新冠疫苗接受意愿与信心的**变化趋势未被研究**

本研究的目的

1）了解**100+国家**新冠疫苗犹豫（及其影响因素）的**流行状况与时间趋势**

2）各国家、区域**新冠疫苗接受度与信心变化的潜在诱因/对策**

方法学

1）**爬取约700万条新冠疫苗相关推特（全样本）**
2）利用课题组背靠背人工标记的超过8000条推特训练基于**CT-BERT**的深度学习模型，(多标签)分类推特文本
3）用各标签在（特定时间、地点的）全部推特的占比来评估公众态度

基于疫苗犹豫模型的分析框架

WHO：疫苗犹豫“3C”模型

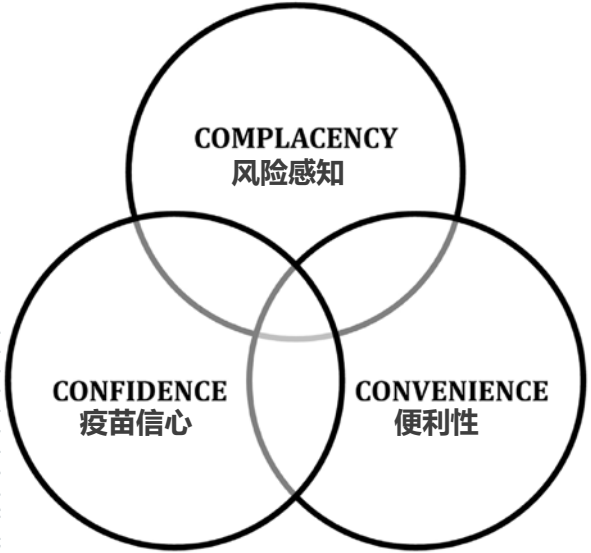


Fig. 2. “Three Cs” model of vaccine hesitancy.

(MacDonald N et al. 2015 *Vaccine*)

社交媒体数据的编码框架

- 该推特是否表达了疫苗犹豫?
  - 是/否
- 疫苗信心?（Confidence）
  - 新冠疫苗是否重要?
  - 新冠疫苗是否有效?
  - 新冠疫苗是否安全?
  - 是否信任政府?
  - 是否信任专家?
- 新冠风险感知(Complacency)
- 便利性（Convenience）？
  - 是否能买到？（可及性）
  - 是否买得起？（可支付性）
- 信息环境、疫苗品种等

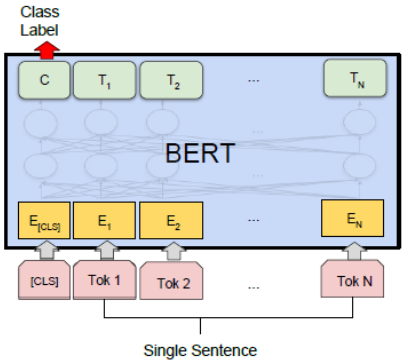
(Hou Z et al. 2021 *JMIR preprint*)

深度学习：人工编码框架中精度足够的类别

训练出的深度学习模型可以自动识别推特中是否传达：

- 疫苗犹豫
  - 是/否
- 疫苗信心 (Confidence)
  - 有效
  - 不安全
  - 不信任政府
- 负面信息
- 对疫苗上市的信心

(Zhou X et al. 2021 *MEDRXIV*)



(b) Single Sentence Classification Tasks: SST-2, CoLA

System	MNLI-(m/mm) 392k	QQP 363k	QNLI 108k	SST-2 67k	CoLA 8.5k	STS-B 5.7k	MRPC 3.5k	RTE 2.5k	Average -
Pre-OpenAI SOTA	80.6/80.1	66.1	82.3	93.2	35.0	81.0	86.0	61.7	74.0
BiLSTM+ELMo+Attn	76.4/76.1	64.8	79.8	90.4	36.0	73.3	84.9	56.8	71.0
OpenAI GPT	82.1/81.4	70.3	87.4	91.3	45.4	80.0	82.3	56.0	75.1
$BERT_{BASE}$	84.6/83.4	71.2	90.5	93.5	52.1	85.8	88.9	66.4	79.6
$BERT_{LARGE}$	**86.7/85.9**	**72.1**	**92.7**	**94.9**	**60.5**	**86.5**	**89.3**	**70.1**	**82.1**

Dataset	BERT-LARGE	CT-BERT	ΔMP
COVID-19 Category (CC)	0.931	0.949	25.88%
Vaccine Sentiment (VC)	0.824	0.869	25.27%
Maternal Vaccine Stance (MVS)	0.696	0.748	17.07%
Stanford Sentiment Treebank 2 (SST-2)	0.937	0.944	10.67%
Twitter Sentiment SemEval (SE)	0.620	0.654	8.97%
Average	0.802	0.833	17.57%

关于CT-BERT模型

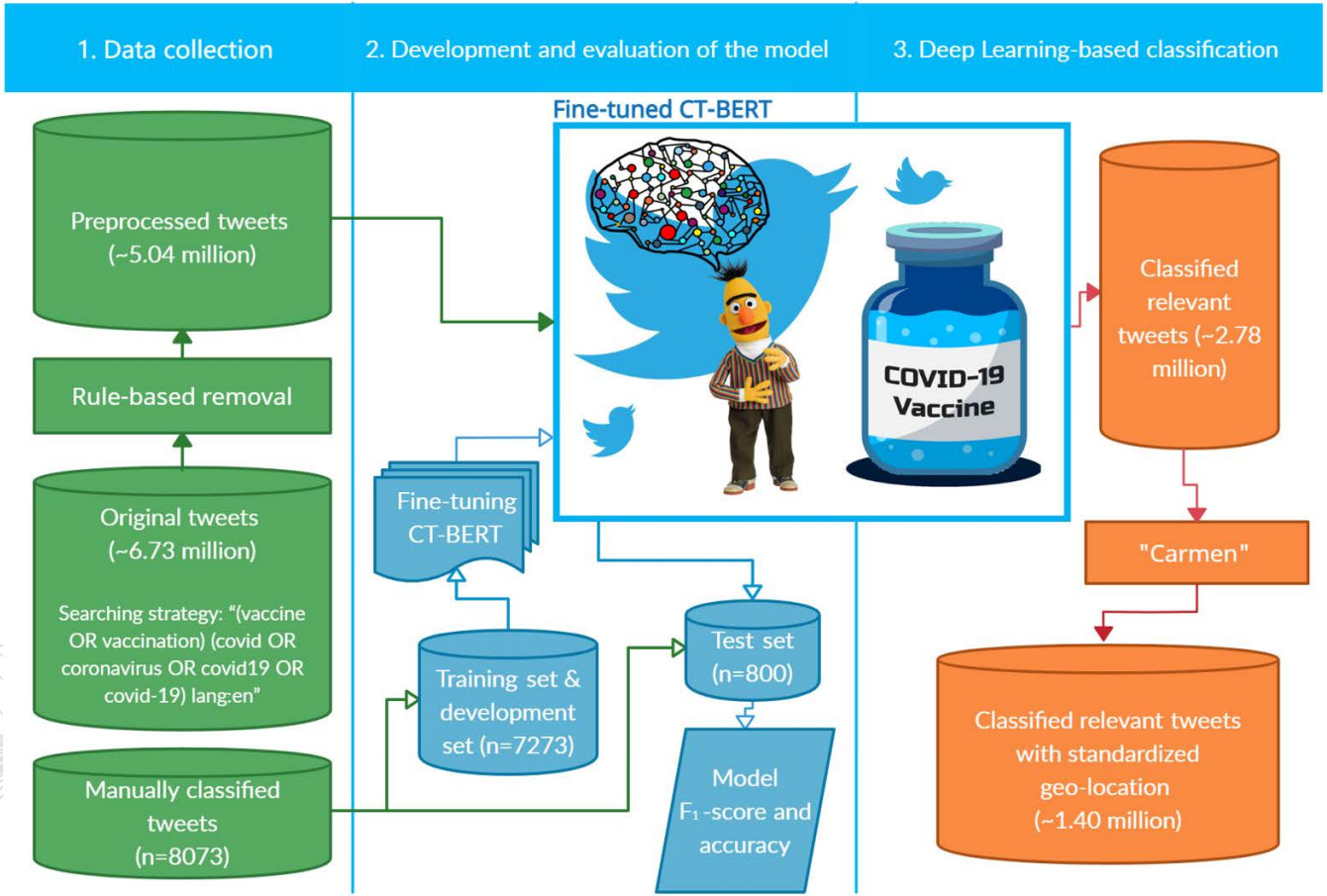
COVID-Twitter-BERT (CT-BERT)的原型是Bidirectional Encoder Representations from Transformers(**BERT**) (**2018**, 18000+引用).

BERT是采用了Transformers (2017, 20000+引用)注意力机制的12/24层神经网络，并在数十亿语料中预训练，使得其在General Language Understanding Evaluation benchmark所有测试项目中一致超越先前所有自然语言处理模型，相比之前最好的模型提升了**7%**的绝对精度

CT-BERT (2020) 采用了与BERT相似的预训练机制，进一步用新冠相关的1.6亿条推特预训练24层版BERT，精度提升10-30%。

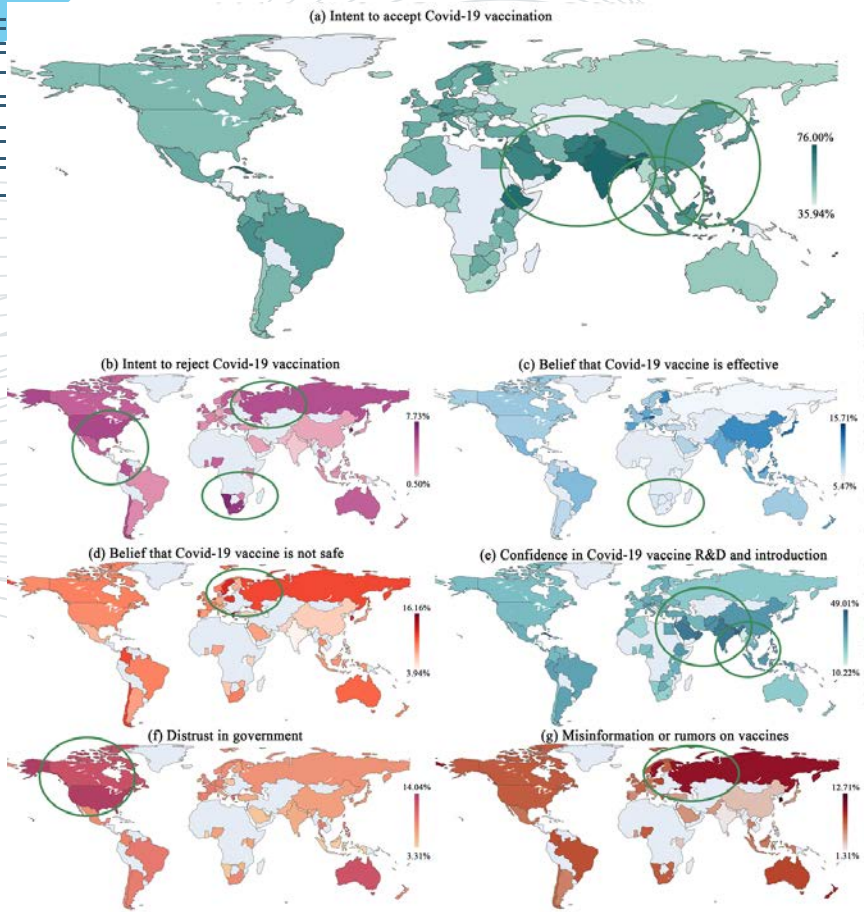
Ashish Vaswani et al. (2017); Jacob Devlin et al.(2018); Martin Müller et al. (2020)

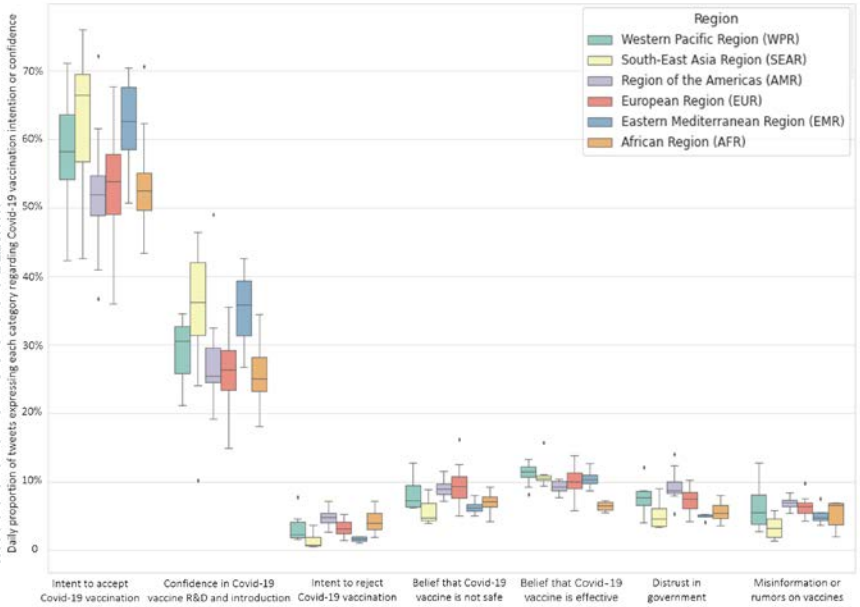
数据处理过程概览


1. Data collection
2. Development and evaluation of the model
3. Deep Learning-based classification
Preprocessed tweets (~5.04 million)
Rule-based removal
Original tweets (~6.73 million)
Searching strategy: "(vaccine OR vaccination) (covid OR coronavirus OR covid19 OR covid-19) lang:en"
Manually classified tweets (n=8073)
Fine-tuned CT-BERT
COVID-19 Vaccine
Fine-tuning CT-BERT
Training set & development set (n=7273)
Test set (n=800)
Model F1-score and accuracy
Classified relevant tweets (~2.78 million)
"Carmen"
Classified relevant tweets with standardized geo-location (~1.40 million)

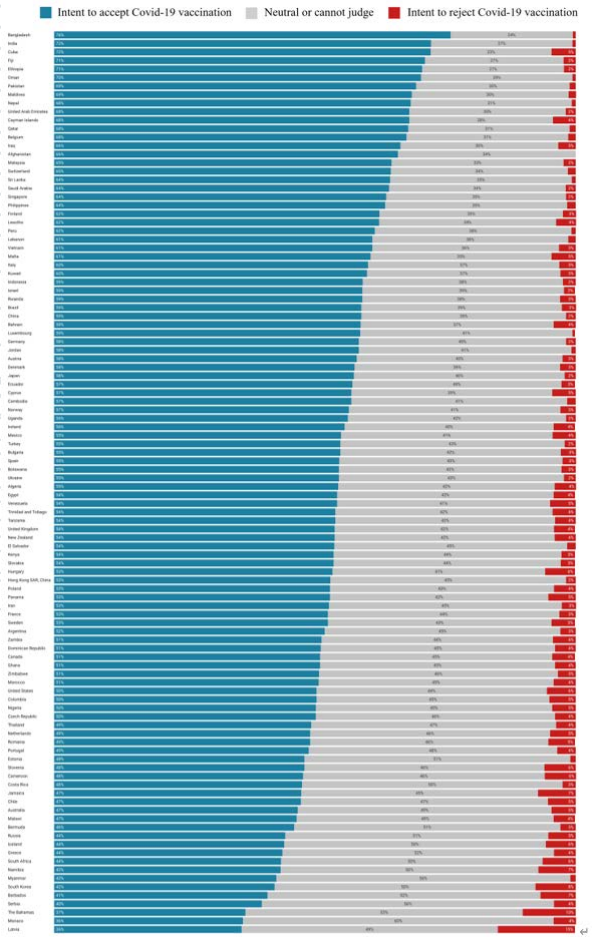
结果部分

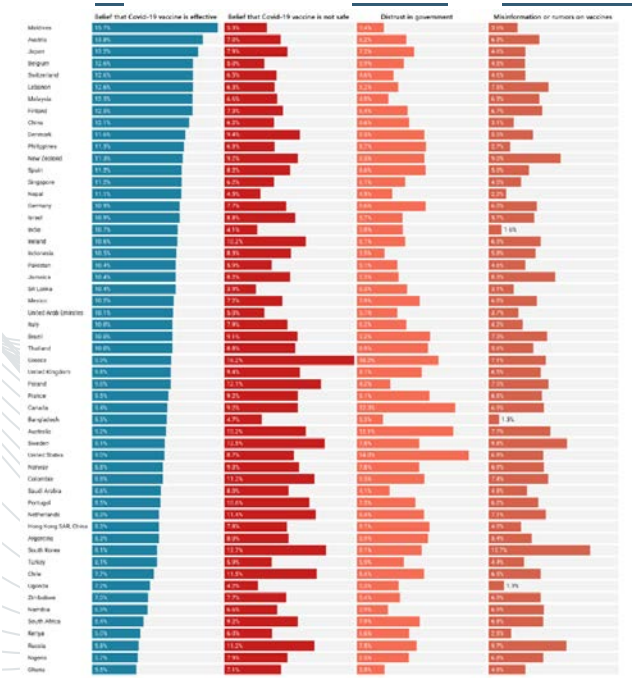
全球新冠疫苗接受度/信心分布（全球地图）

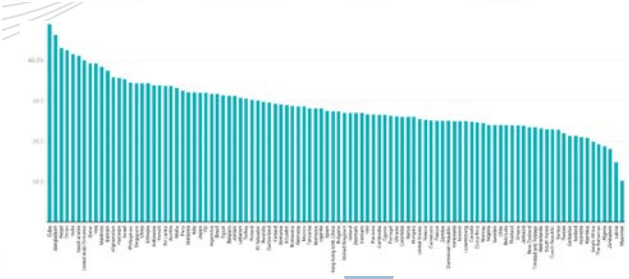




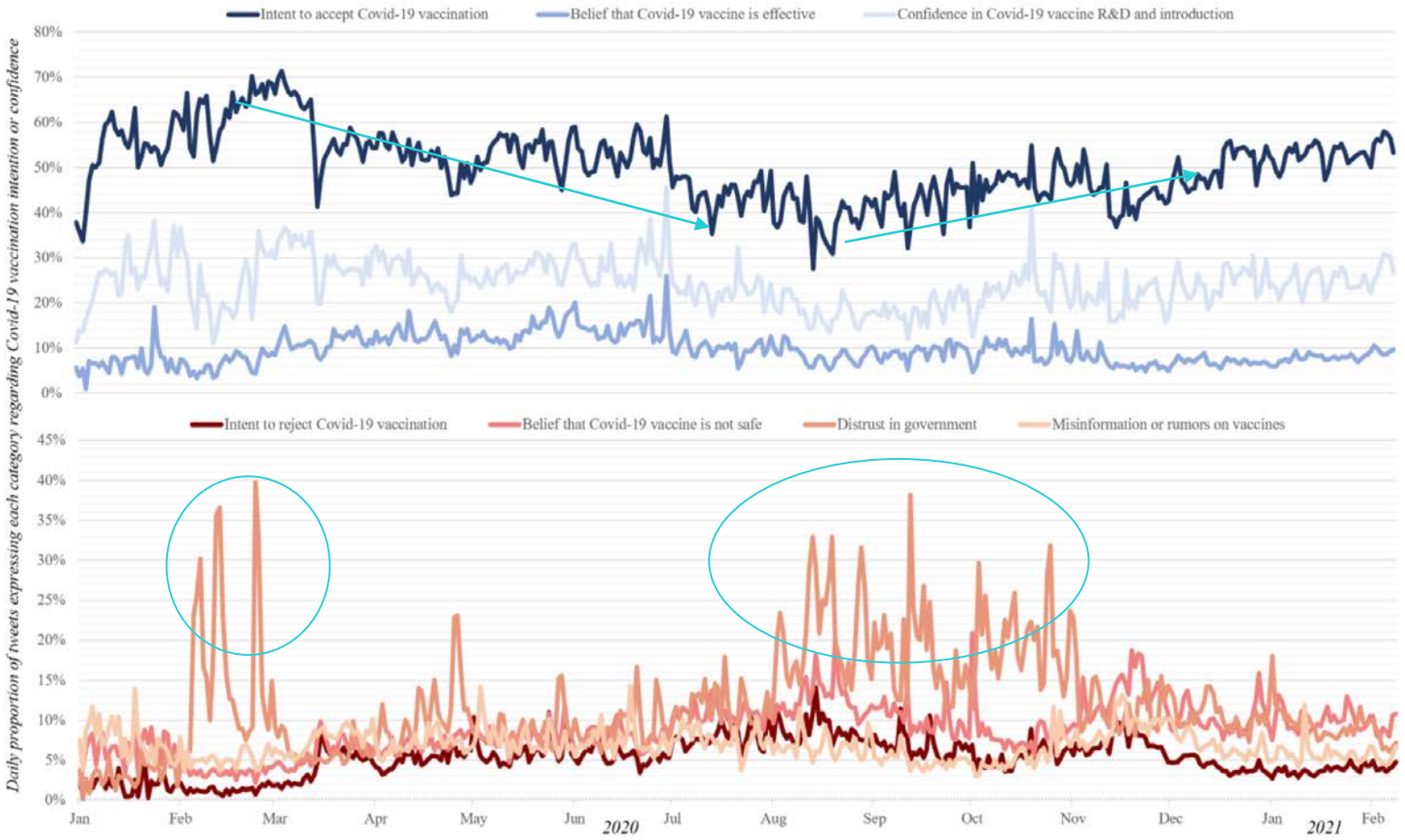
各国新冠疫苗接受度/信心分布（详细数据）



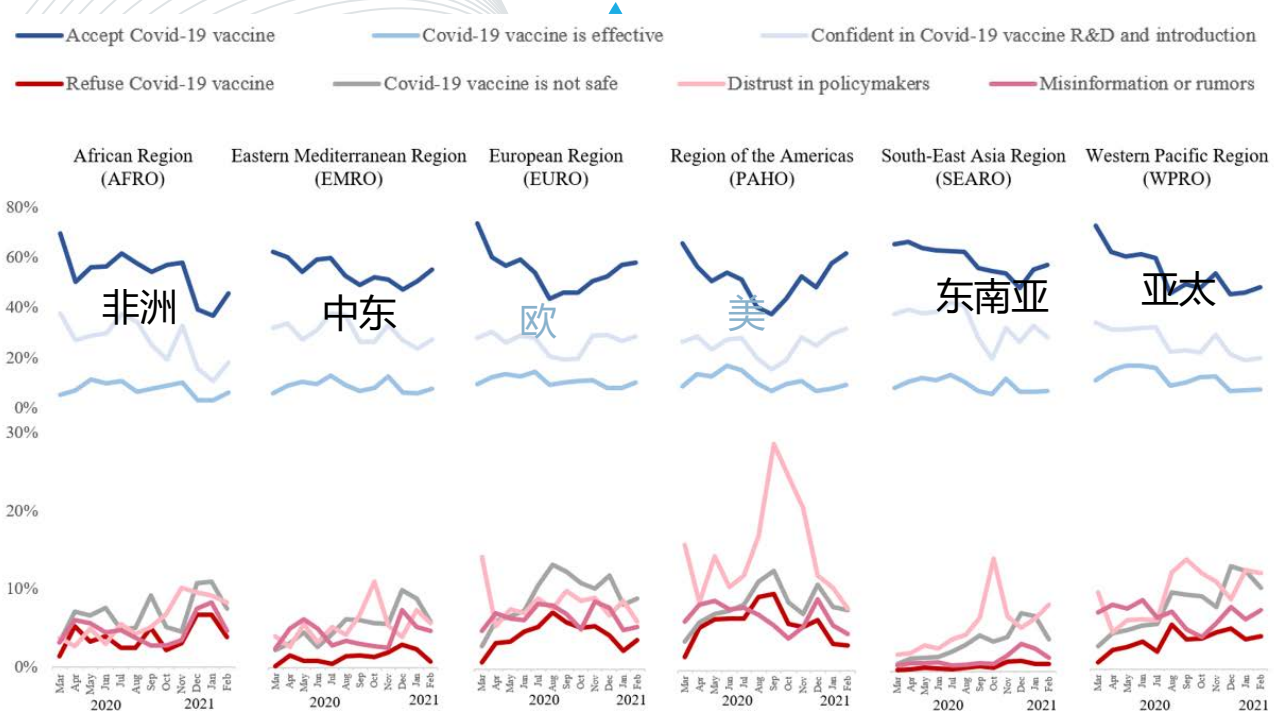




全球新冠疫苗接受度与信心变化



全球（按区域）新冠疫苗接受度变化

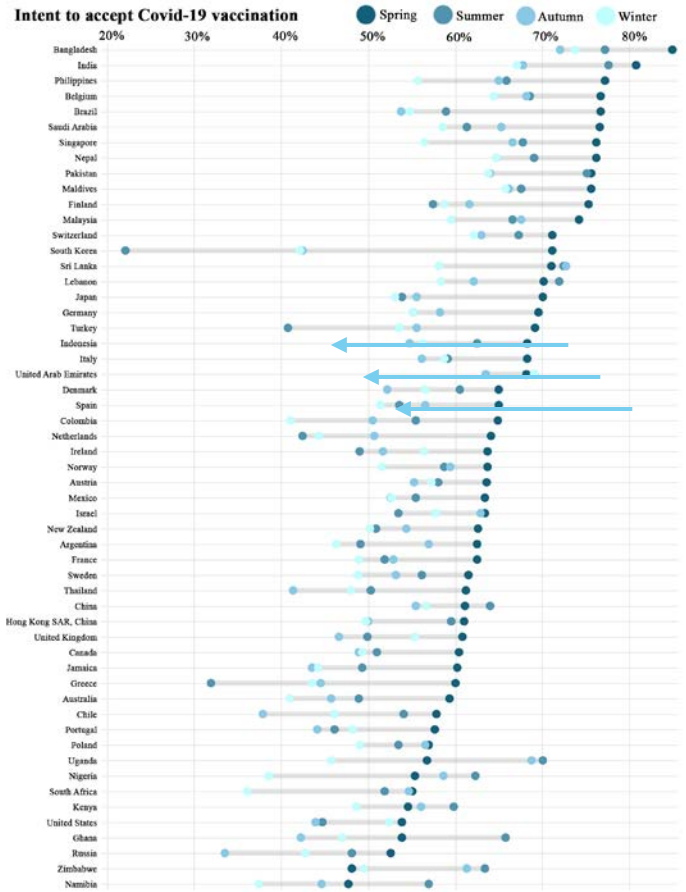


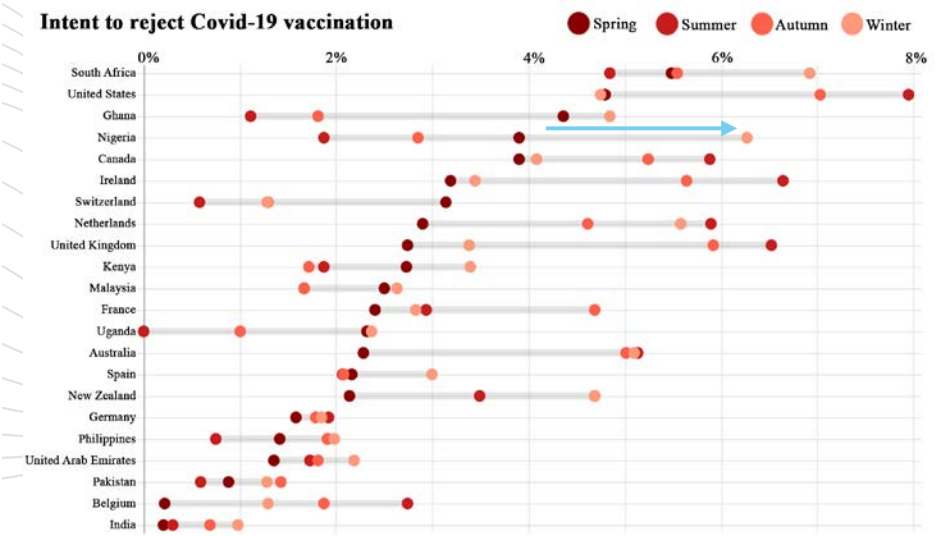
各国新冠疫苗接受度按季节变化

Spring：2020.3-5，Summer：2020.6-8，Autumn：2020.9-11，Winter：2020.12-2021.2

深色→2020年初

浅色→今年年初

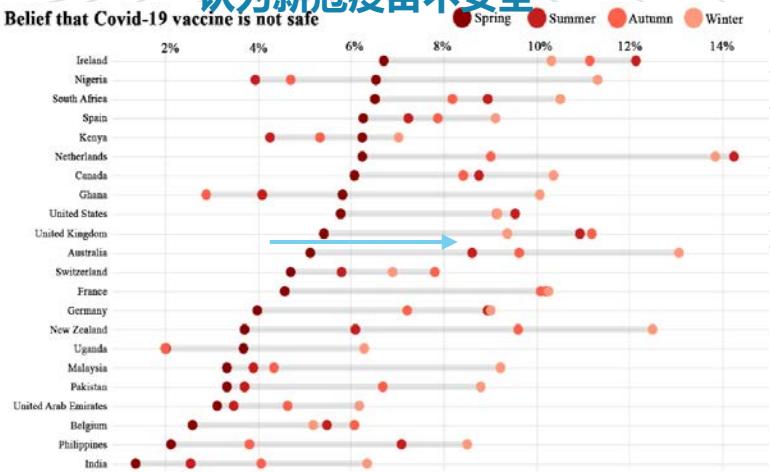




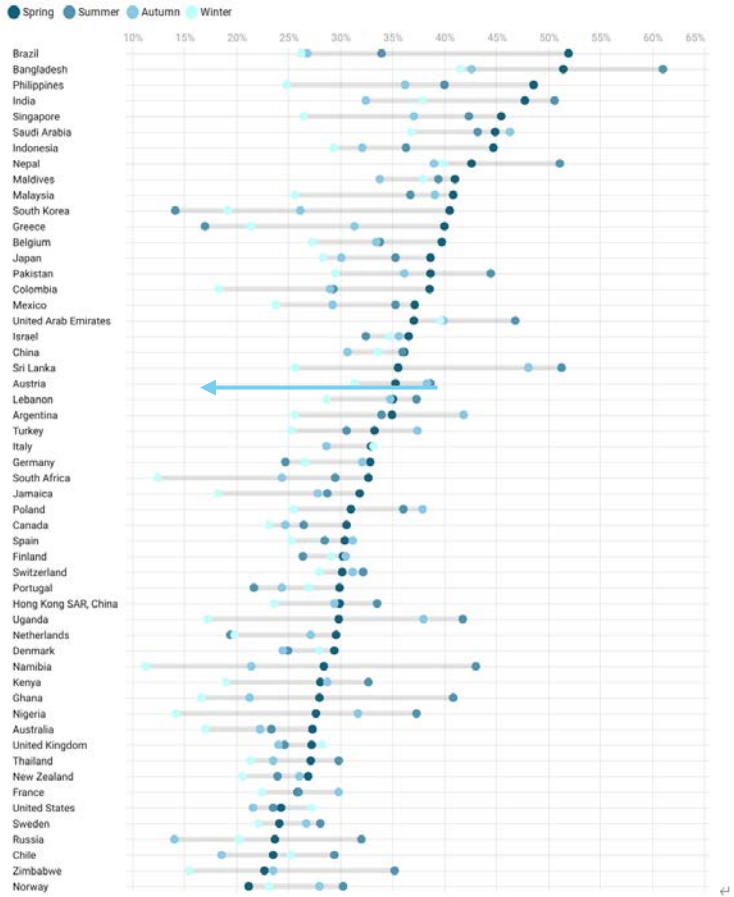
各国疫苗信心按季节变化

认为新冠疫苗有效

认为新冠疫苗不安全

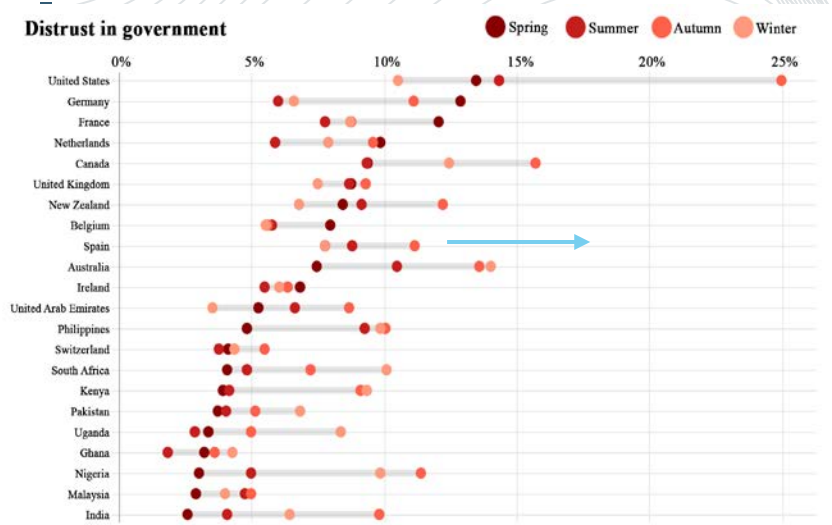


对新冠疫苗研发上市的信心

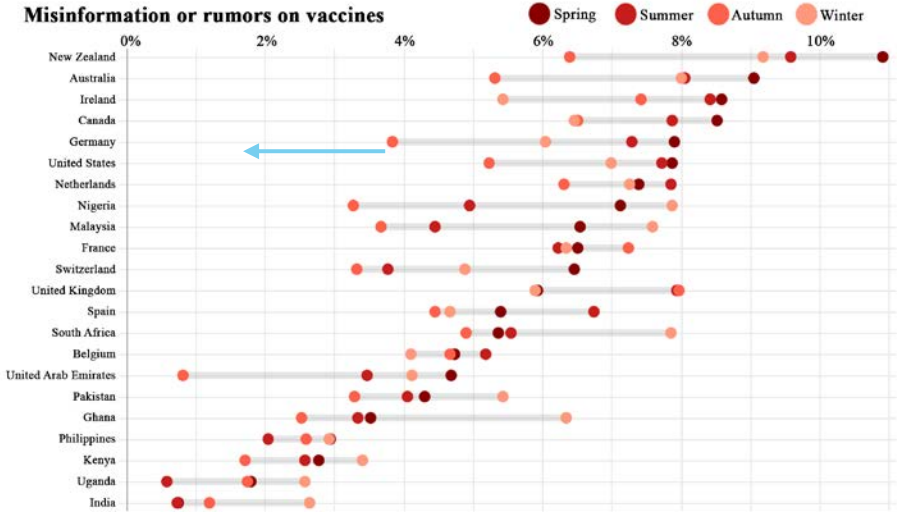


谣言/政府信任度按季节变化

不信任政府的比例



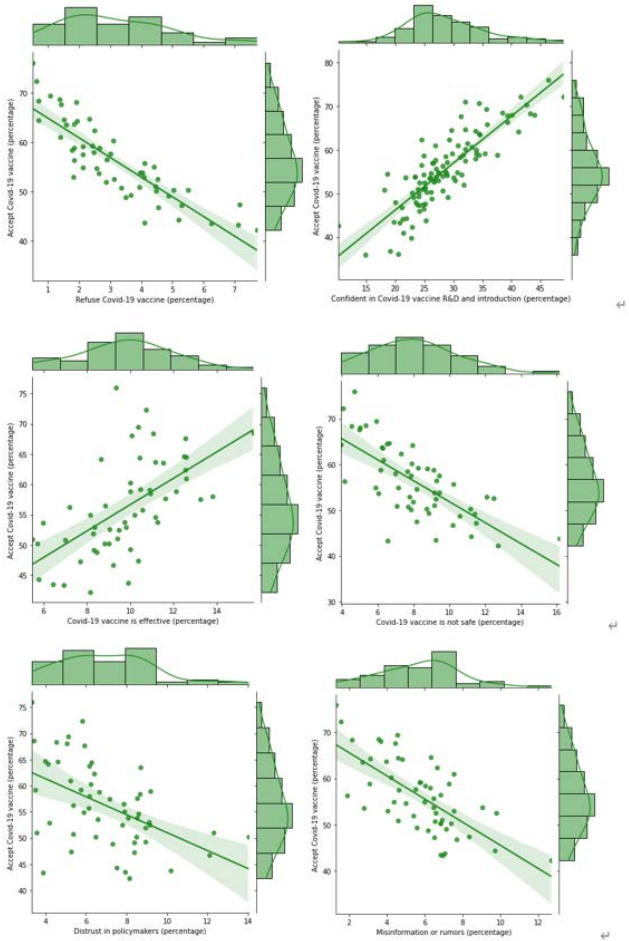
谣言/不实信息



线性回归的结果:

认为疫苗不安全
谣言、不信任政府广泛分布的国家
→疫苗犹豫

认为疫苗有效、对疫苗上市研发有信心的国家
→接受疫苗接种



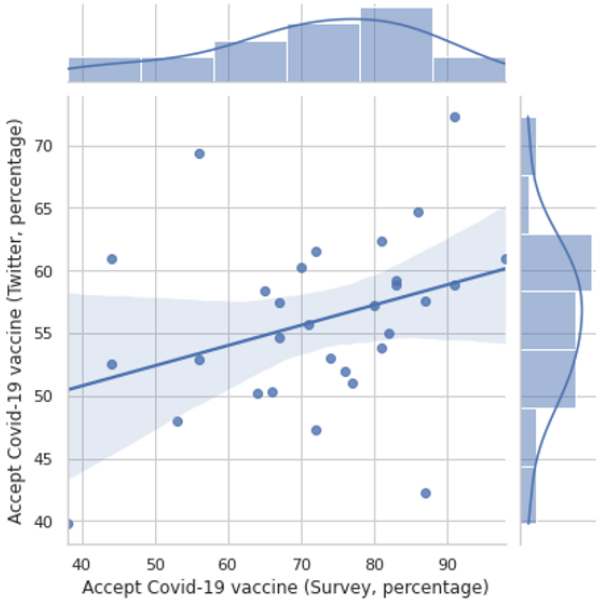
公众参与度

	Follower	Retweet	Favorite	Reply	Quote
All tweets	2502.85(20737.55)	2.55(194.20)	12.94(1307.94)	0.91(23.40)	0.28(13.84)
Intent to accept Covid-19 vaccination	2511.37(21878.67)	2.44(186.15)	15.24(1636.72)	0.84(13.87)	0.24(14.99)
Intent to reject Covid-19 vaccination	2224.92(16224.20)	2.26(82.22)	8.40(310.48)	1.45(81.98)	0.41(15.03)
Belief that Covid-19 vaccine is effective	2583.09(22342.67)	1.66(87.84)	11.03(708.83)	0.81(9.59)	0.18(12.44)
Belief that Covid-19 vaccine is unsafe	2137.65(12929.20)	3.07(210.00)	11.99(1052.35)	0.87(7.98)	0.34(12.26)
Confidence in Covid-19 vaccine R&D and introduction	2511.45(21297.92)	1.97(154.43)	14.09(968.68)	0.80(11.11)	0.22(9.11)
Distrust in government	2891.09(20814.53)	3.26(155.91)	11.77(664.82)	0.87(15.83)	0.30(10.28)
Misinformation or rumors on vaccines	2293.98(20954.31)	3.00(173.09)	17.98(1228.41)	1.16(20.42)	0.37(17.88)

Mean (Standard Deviation) are shown.

负面消息（尤其是谣言）公众参与度更高

新冠疫苗接受度分布稳健性——与现有问卷研究的比较



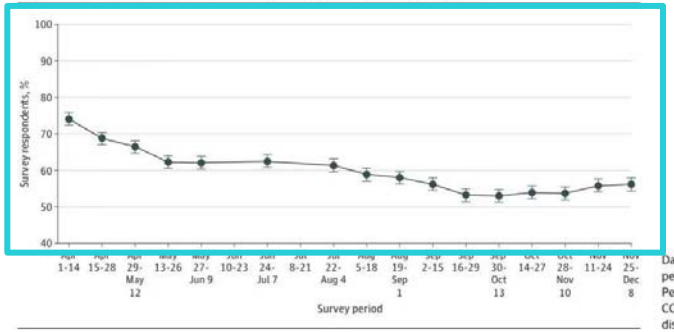
Olivier J Wouters et al.
(2021, *Lancet*)

Wouters OJ, Shadlen KC, Salcher-Konrad M, Pollard AJ, Larson HJ, Teerawattananon Y, Jit M. Challenges in ensuring global access to COVID-19 vaccines: production, affordability, allocation, and deployment. The Lancet. 2021 Feb 12.

P=0.06, R2=0.12

时间趋势稳健性——与现有问卷研究的比较

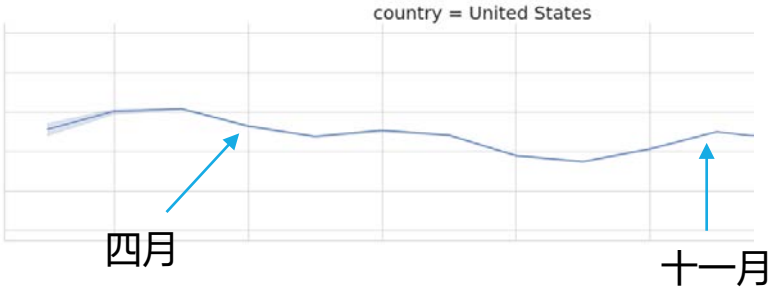
Figure. Percentage of US Adults Who Say They Are Likely to Get a COVID-19 Vaccine



Data were not collected for the periods of June 10-23 and July 8-21. Percentages and 95% CIs are plotted. COVID-19 indicates coronavirus disease 2019.

Peter G. Szilagyi et al. (2021, *JAMA*)

United States (Twitter)



本研究的优势

01. 迄今**最大**（106国）的全球新冠疫苗信心与接受度比较研究——
为被既往研究忽视的数十国家提供疫苗犹豫数据

02. 迄今**唯一多国和多区域区域疫苗接受度与信心变化的研究**，可作为政策施行效果和未来相关研究的参考；
迄今**唯一**关于**新冠疫苗相关不信任政府与疫苗谣言在各国、各区域流行状况的研究**

03. 社交媒体数据丰富；全样本

04. 使用了性能最好的深度学习模型之一，达到了超越人工分析的精度

对未来研究的展望

多语言

训练多语言模型（如ERNIE-M），挖掘来自几十或上百种语言的社交媒体数据

多社交媒体

各国常用的社交媒体，如微博除推特外，国外对脸书的研究也较为丰富。

实时监测

制作网页，实时呈现各国新冠疫苗接受度与信心

感谢聆听！

周新宇
复旦大学公共卫生学院本科生
xinyuzhou17@fudan.edu.cn